

NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE

PH1012 QUIZ 1 SEMESTER 1, 2023-2024

16th September 2023, 1:45 pm – 3:15 PM

Constants List

Gravitational Acceleration	g	9.8 m/s^2
Mass of Earth	M_e	$5.97 \times 10^{24} \text{ kg}$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$
Avogadro's number	N_A	$6.02 \times 10^{23} \text{ molecules/mol}$
Universal gas constant	R	8.314 J/mol K
Absolute zero	0 K	-273.15°C
Pressure conversion	1 bar	$10^5 \text{ N/m}^2 \text{ (or Pa)}$
	1 atm	$1.01 \times 10^5 \text{ Pa}$

Tutorial Group	Questions	Short Q	Long Q1	Long Q2	Long Q3	Total
	Score	/ 25	/ 10	/ 15	/ 10	/ 60

Basics and Thermal Physics

$$\Delta L = \alpha L_i \Delta T, \quad \Delta V = \beta V_i \Delta T, \quad PV = nRT, \quad Q = mc\Delta T, \quad Q = ml$$

$$T = \frac{P(T) - P(0)}{P(100) - P(0)} \times 100, \quad T = 273.16 \text{ K } \lim_{P_{tr} \rightarrow 0} \frac{P}{P_{tr}},$$

Linear Motion

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}_f - \vec{r}_i}{t_f - t_i}, \quad \vec{v}_{int} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}, \quad v = \frac{dx}{dt},$$

$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}, \quad \vec{a}_{int} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{r}}{dt^2}, \quad a = \frac{dv}{dt}$$

$$v_f = v_i + at, \quad x_f = x_i + \frac{1}{2}(v_i + v_f)t, \quad v_f^2 = v_i^2 + 2a(x_f - x_i), \quad x_f = x_i + v_i t + \frac{1}{2}at^2$$

$$y = (\tan \theta_i)x - \left(\frac{g}{2v_i^2 \cos^2 \theta_i} \right) x^2 \quad (\text{taking downwards as } -ve, \quad \theta_i \text{ is measured above } +ve \text{ } x \text{ axis}),$$

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g}, \quad R = \frac{v_i^2 \sin 2\theta_i}{g}$$

$$\Sigma \vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}, \quad \vec{p} = m\vec{v}, \quad f_k = \mu_k N, \quad f_s \leq \mu_s N \quad |F_{sp}| = kx \quad v = \frac{mg}{b} (1 - e^{-\frac{b}{m}t})$$

Rotational Motion

$$a_c = \frac{v^2}{r} = r\omega^2, \quad a_t = \frac{d|\vec{v}|}{dt}, \quad s = r\theta, \quad \omega = \frac{d\theta}{dt} = \frac{v}{r}, \quad \alpha = \frac{d\omega}{dt} = \frac{a}{r}; \quad \omega_f = \omega_i + \alpha t,$$

$$\theta_f = \theta_i + \frac{1}{2}(\omega_i + \omega_f)t, \quad \omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i), \quad \theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2$$

Mathematics

$$\tan A = \frac{\sin A}{\cos A}; \quad \sin^2 A + \cos^2 A = 1; \quad \sec^2 A = \frac{1}{\cos^2 A} = \tan^2 A + 1; \quad x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{where } ax^2 + bx + c = 0;$$
$$\frac{d}{dx}(ax^n) = anx^{n-1}; \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1); \quad \int \frac{dx}{x} = \ln x$$

Instructions:

a) Answer all questions.

b) Working for all questions must be shown clearly in the space provided.

Short Questions (25 points)

1. [3 points] The “per square foot” price of a 5 bedroom flat is listed as \$542 / ft². Calculate the price if the 5 bedroom flat has a floor area of 120 m². You are given that 1 m = 3.281 ft.

ANS: _____

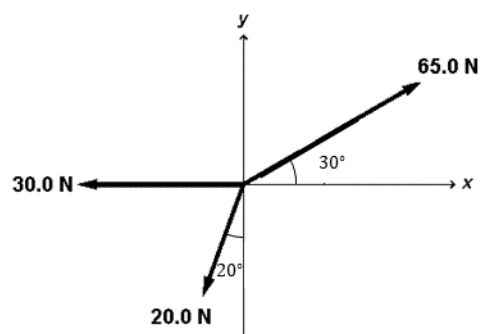
2. [3 points] A boy got a balloon at the airport and brought it on an airplane. At the airport, the temperature was 30°C and the volume of balloon was 0.50 m^3 . The ambient pressure is 1.00 atm . Cruising at 10 km high, the temperature inside the plane is 18°C and the pressure is 0.80 atm . Determine the new volume of the balloon in the plane.

ANS: _____

3. [4 points] A car traveling along a straight road increases its speed from 20.0 m/s to 40.0 m/s in a distance of 180 m . If the acceleration is constant, how much time elapses while the car moves this distance?

ANS: _____

4. [4 points] Referring to the diagram, determine the magnitude of the resultant of these three forces.



ANS: _____

5. [3 points] Which of the following is / are not at constant acceleration?
(Ignore air resistance and friction.)

- A. The moment after a rock is thrown off the top of a building to just before it hits the ground.
- B. An elevator moving from rest from the ground floor and stopping on the fifth floor.
- C. Both A and B.
- D. None of the above.

ANS: _____

6. [3 points] Your little sister wants you to give her a ride on her sled on snow. On level ground, if the same force F is used, which method is the frictional force smaller?



- A. Pushing her from behind.
- B. Pulling her from the front.
- C. It is the same in both cases.
- D. Not enough information.
- E. It is impossible to move the sled this way.

ANS: _____

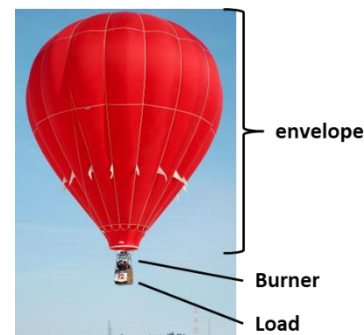
7. [5 points] A child tosses a ball vertically upwards. Its total time in the air is T and maximum height is H . What is its height (expressed as a fraction of H) after it has been in the air for a time $T/4$? Ignore air resistance. Leave your answer as a fraction of H .

ANS: _____

Structured Questions (35 points)

1. [10 points] The diagram shows a simple diagram of the hot-air balloon. A hot air balloon consists of a bag, called an envelope, which contains heated air. Suspended beneath is the load which comprises the basket carrying equipment, passengers and a burner (an open flame produced by burning liquid propane). The variation of density of air with temperature inside the envelope is given in the table below. You can use $g = 9.8 \text{ N/kg}$.

Temperature of Air ($^{\circ}\text{C}$)	0	20	80	100	120
Density of Air (kg/m^3)	1.292	1.204	0.9996	0.9461	0.8978



- a. Using the information given in the table, calculate the upthrust per m^3 U and the weight of air per m^3 W_A , when the surrounding air temperature is 20°C with the balloon air temperature at 100°C . Show that lifting force per unit volume (m^3) $L (= U - W_A)$ is about 2.5 N/m^3 .

ANS: $U =$ _____

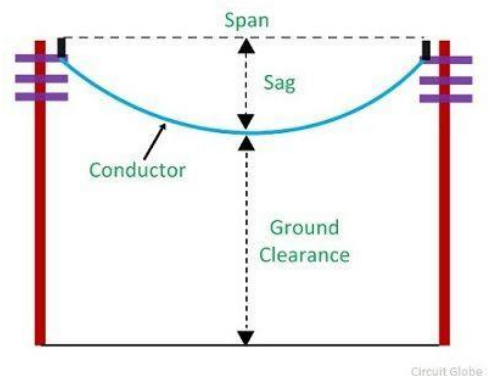
$W_A =$ _____

- b. If the volume of the envelope is fixed 1800 m^3 and the surrounding air temperature is 20°C with the balloon air temperature at 100°C , determine the acceleration of the hot - air balloon system if total mass of the load and the skin of the balloon is 320 kg .

ANS: _____

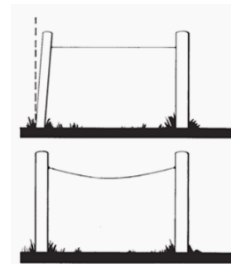
- c. In reality, the actual acceleration when the balloon is ascending is lesser than the answer you calculated in part (b). Explain why this is so.

2. [15 points] There are a number of factors to consider when calculating the sag of overhead electrical power transmission cables. These include weight, weather and thermal aspects of the conductor cables. In a certain set-up, the span of the two towers supporting the cables is 80.0 m and the conducting cable is made of aluminum with 1.2 cm diameter. The temperature in this region range between -50°C and $+40^{\circ}\text{C}$.



The density of aluminum is $\rho = 2.7 \text{ g / cm}^3$. The coefficient of linear expansion of aluminum is $\alpha = 25 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$ and its specific heat capacity is $900 \text{ J/kg}^{\circ}\text{C}$.

- a. If a cable was strung without sag, it would snap during cold weather. Determine the minimum extra length of the aluminum cable (at 20°C) needed so as to avoid the cable from snapping. You can assume that the span is constant at 80.0 m for all temperatures.



ANS: _____

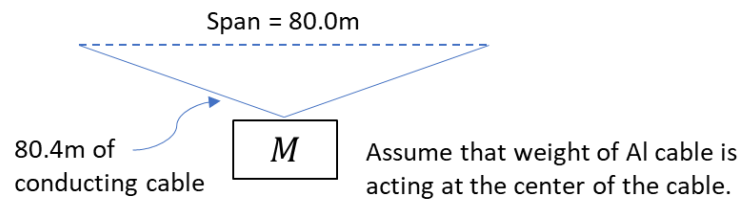
- b. If 80.4 m of 1.2 cm diameter aluminium cable is used, determine the mass M of the cable.

ANS: _____

- c. If the power lost as heat in the 80.4 m of cable is 470 J / s , determine the increase in temperature per minute. Assume no heat is lost to the surrounding.

ANS: _____

- d. The tension (T) in the cable can be estimated in a very simplified way by assuming that the weight (Mg) of the aluminium cable just acts in the center of cable and that the rest of the cable is massless. With the above assumptions, determine the tension in the cable.

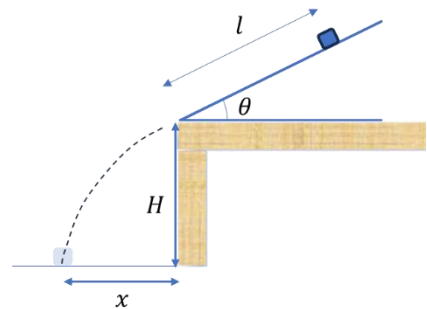


ANS: _____

- e. State explicitly two weather conditions that will increase the tension in the overhead transmission cables.

1. _____
2. _____

3. [10 points] A block is released from rest and slides down a frictionless inclined plane at an angle θ . The inclined plane is on a platform which has a height $H = 0.90\text{ m}$ above the ground. The block lands at a horizontal distance $x = 0.48\text{ m}$ from the base of the platform. The time that the block leaves the base of the inclined plane till it just lands on the ground is accurately timed as $t = 0.42\text{ s}$. You can ignore air resistance.



- a. Determine the horizontal velocity v_x of the block.

ANS: _____

- b. Determine the angle θ .

ANS: _____

- c. Determine the length l along the frictionless inclined plane that the block has slide.

ANS: _____

---- The End ----